

Advanced Data Analysis and Machine Learning (Project 1a)

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Communication

We have established a WhatsApp group for communication among the project members and decided to occasionally meet in person on campus. The code is shared using a GitHub repository — <https://github.com/nedelamatej/adaml-kul>.

Exploratory Analysis

The *NASA Turbofan Jet Engine* dataset consists of four subsets (FD001 – FD004) each with a training set, a test set and a vector of true remaining useful life (RUL) values, distinguished by operating conditions and fault modes, as shown in Table 1. Training data were loaded into Python as four matrices (20631, 53759, 24720, 61249 observations as rows and 26 predictors as columns). All further exploratory analysis in this report focuses on FD001 (single condition, single fault mode) for clarity. However, the same steps will also be applied to the other subsets.

Subset	Operating condition	Fault modes	Units	Rows
FD001	Single (sea level)	Single (HPC degradation)	100	20631
FD002	Six (discrete)	Single (HPC degradation)	260	53759
FD003	Single (sea level)	Two (HPC degradation and fan degradation)	100	24720
FD004	Six (discrete)	Two (HPC degradation and fan degradation)	249	61249

Table 1: *NASA Turbofan Jet Engine* dataset subsets (all of them) and their characteristics.

Each subset has 26 columns, starting with unit number, time (cycle), 3 operational settings and 21 sensor measurements, as shown in Table 3, with symbols and physical meaning taken from *Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation* (Saxena et al., 2008). The 3 operational settings (altitude, Mach number, TRA) take only 6 discrete combinations and could be treated as a single categorical attribute for operating condition.

It has been verified that there are no missing values and that each unit's time series is continuous without any gaps. However, the units have varying lifetimes, as shown in histogram in Figure 1. Columns T2, P2, Nf_dmd and PCNfR_dmd have zero standard deviation within each operating condition and therefore are redundant for modelling. Other sensor scales differ by orders of magnitude and require normalization or standardization before modelling.

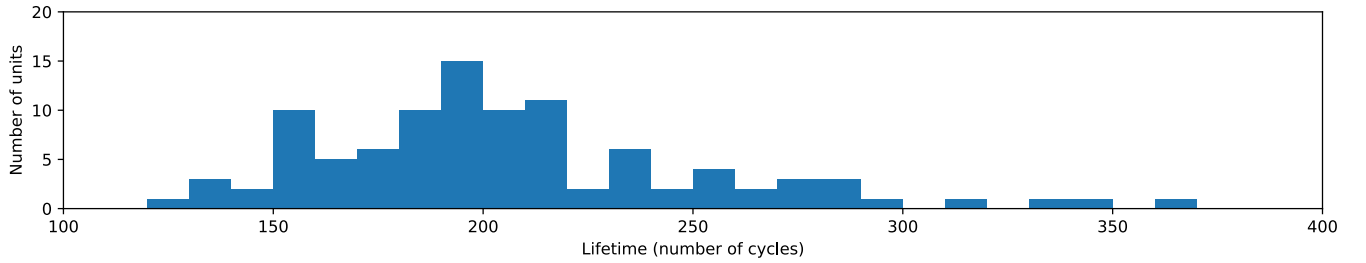


Figure 1: Histogram of unit lifetimes for FD001.

Given the two fault modes (HPC and fan degradation), sensors expected to be most informative for RUL include T30, P30, Ps30 (HPC outlet conditions), Nf and NRf (fan speed). Summary statistics and plots for the FD001 HPC outlet sensors are shown in Table 2 and Figure 2. It can be observed that the total temperature and static pressure at HPC outlet rise over the unit's lifetime.

Symbol	Description	Unit	Mean	Std. dev.	Min.	Max.
T30	Total temperature at HPC outlet	°R	1590.52	6.13	1571.04	1616.91
P30	Total pressure at HPC outlet	psia	553.37	0.89	549.85	556.06
Ps30	Static pressure at HPC outlet	psia	47.54	0.27	46.85	48.53

Table 2: *NASA Turbofan Jet Engine* dataset predictors (three of them) and their statistics.

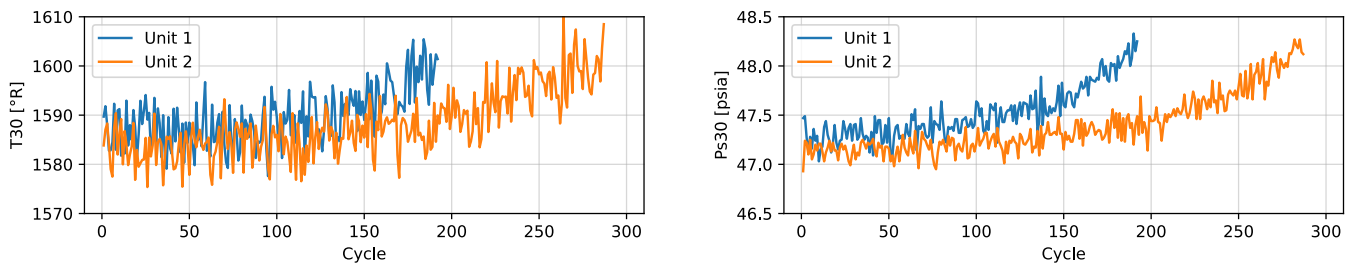


Figure 2: Total temperature (T30) and static pressure (Ps30) at HPC outlet for FD001.

Principal Component Analysis

The principal component analysis was applied to all of the four subsets. Before applying the principal component analysis on the dataset, all sensor features were standardized to eliminate the scale difference and to ensure equal contribution to the variance calculation. After applying the principal component analysis, we took 2 principal component (PC) which reduced the high-dimensional complexity of the dataset but captured a substantial share of the total variance, though considerably more for FD002 and FD004 (both 97 %) than for FD001 and FD003 (74 % and 76 %, respectively). The high variance captured for FD002 and FD004 likely reflects the added variation from six operating conditions rather than degradation alone, motivating the per-condition normalization planned below.

The biplots (Figure 3) and loading plots (Figure 4) clearly visualize the group of 21 sensors into distinct clusters based on their behaviors telling us which physical parameters (such as temperatures, pressures and speeds) share strong relation. Comparing the subsets highlights how operating environments change engine behavior. Datasets like FD001 and FD003 show smooth and unified degradation trajectories because they operate under constant conditions whereas FD002 and FD004 show multi-branched or scattered clusters due to varying altitude, Mach number and throttle settings. The loading vectors map out physical subsystems behavior, closely aligned sensor arrows (such as specific temperature and pressure pairings) indicating that thermal and aerodynamic subsystems tend to degrade together.

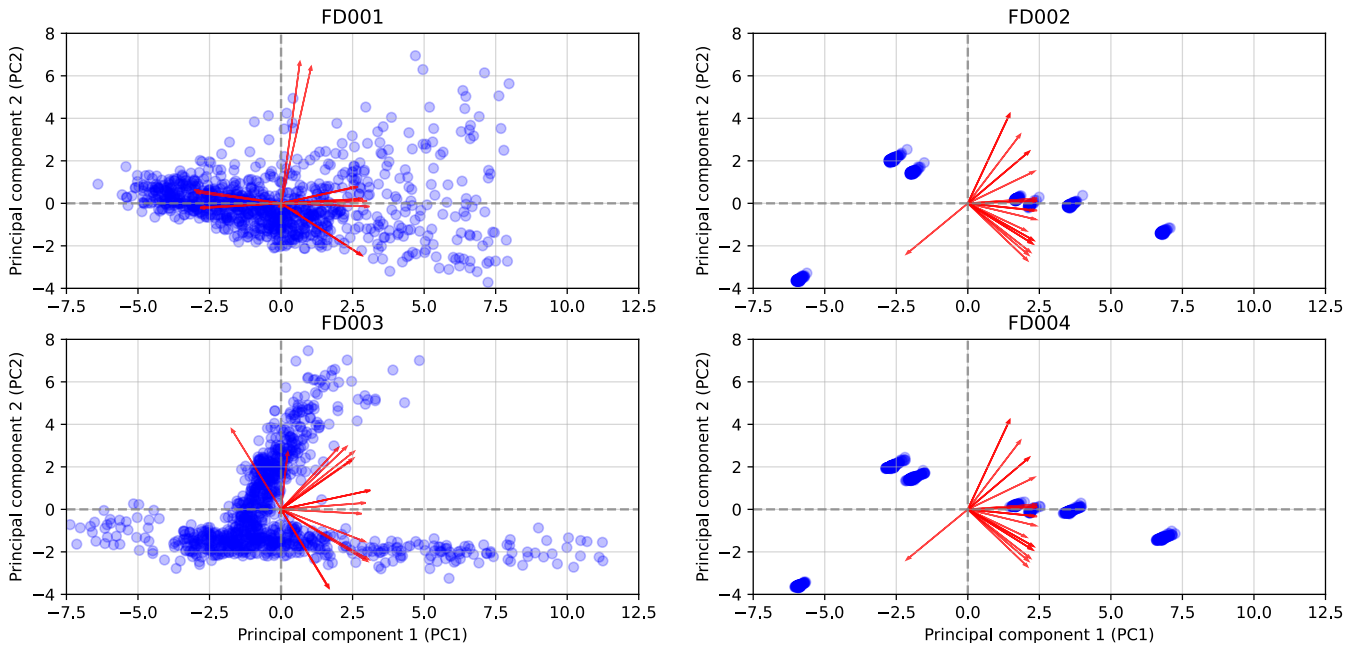


Figure 3: PCA biplots for all NASA Turbofan Jet Engine subsets.

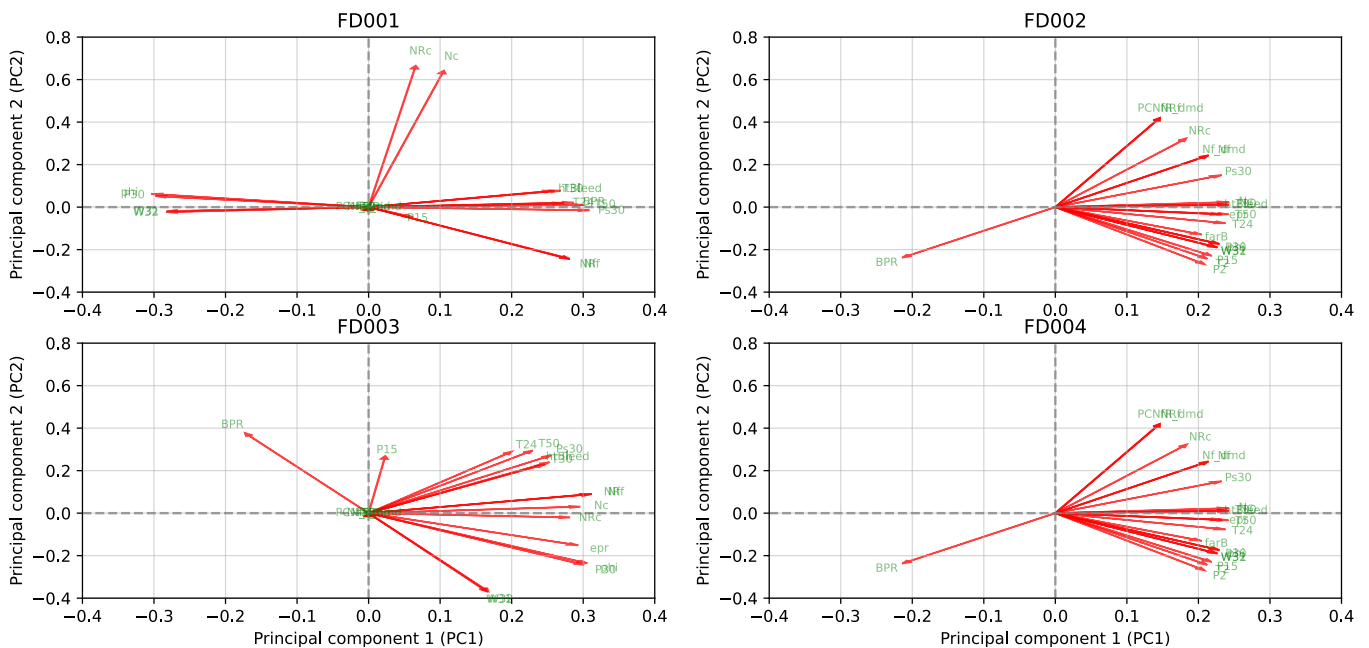


Figure 4: PCA loading plots for all NASA Turbofan Jet Engine subsets.

Pretreatment

All sensor variables will be standardized before further analysis, and the four columns with zero variance (T2, P2, Nf_dmd, PCNfr_dmd) will be removed. No missing or extreme values were found. For FD002 and FD004, sensor readings will additionally be normalized per operating condition before modelling, to separate degradation from changes in operating point.

Appendix

Symbol	Description	Unit
unit_id	Unit number	—
cycle	Time	cycle
altitude	Altitude	K ft
mach	Mach number	—
tra	Throttle resolver angle	—
T2	Total temperature at fan inlet	°R
T24	Total temperature at LPC outlet	°R
T30	Total temperature at HPC outlet	°R
T50	Total temperature at LPT outlet	°R
P2	Pressure at fan inlet	psia
P15	Total pressure in bypass-duct	psia
P30	Total pressure at HPC outlet	psia
Nf	Physical fan speed	rpm
Nc	Physical core speed	rpm
epr	Engine pressure ratio (P50/P2)	—
Ps30	Static pressure at HPC outlet	psia
phi	Ratio of fuel flow to Ps30	pps/psi
NRf	Corrected fan speed	rpm
NRc	Corrected core speed	rpm
BPR	Bypass ratio	—
farB	Burner fuel-air ratio	—
htBleed	Bleed enthalpy	—
Nf_dmd	Demanded fan speed	rpm
PCNfR_dmd	Demanded corrected fan speed	rpm
W31	HPT coolant bleed	lbm/s
W32	LPT coolant bleed	lbm/s

Table 3: *NASA Turbofan Jet Engine* dataset columns (all of them) and their description.